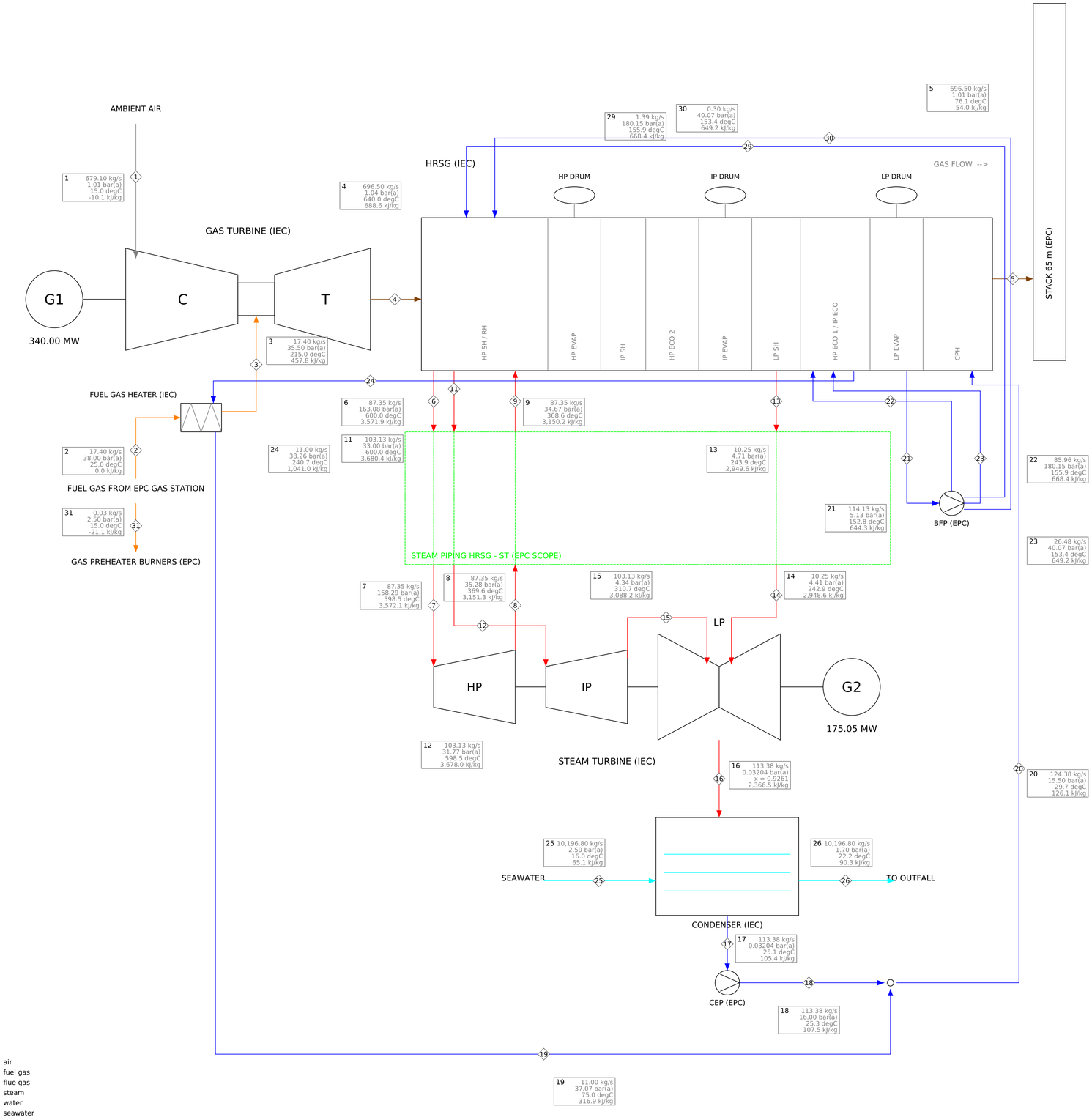


STREAM TABLE - SRC-NG-GT80

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	679.10	1.01	15.0	-10.1	-6.9
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	17.40	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	17.40	35.50	215.0	457.8	8.0
4	GT exhaust gas to HRSG inlet	696.50	1.04	640.0	688.6	479.6
5	Flue gas at HRSG outlet to stack	696.50	1.01	76.1	54.0	37.6
6	HP steam at HRSG outlet (HP SH outlet)	87.35	163.08	600.0	3,571.9	312.0
7	HP steam at ST main stop valve inlet	87.35	158.29	598.5	3,572.1	312.0
8	Cold reheat steam at HP turbine exhaust	87.35	35.28	369.6	3,151.3	275.3
9	Cold reheat steam at HRSG reheater inlet	87.35	34.67	368.6	3,150.2	275.2
10	IP superheated steam (joins cold reheat inside the HRSG)	15.47	35.40	323.1	3,037.3	47.0
11	Hot reheat steam at HRSG outlet	103.13	33.00	600.0	3,680.4	379.5
12	Hot reheat steam at ST IP stop/control valve inlet	103.13	31.77	598.5	3,678.0	379.3
13	LP steam at HRSG outlet	10.25	4.71	243.9	2,949.6	30.2
14	LP steam at ST LP admission	10.25	4.41	242.9	2,948.6	30.2
15	IP turbine exhaust (crossover, internal)	103.13	4.34	310.7	3,088.2	318.5
16	LP turbine exhaust to condenser (UEEP)	113.38	0.0320	x 0.926	2,366.5	268.3
17	Condensate at hotwell outlet / CEP suction	113.38	0.0320	25.1	105.4	11.9
18	Condensate at CEP discharge	113.38	16.00	25.3	107.5	12.2
19	Fuel gas heater water return to condensate	11.00	37.07	75.0	316.9	3.5
20	Condensate to HRSG condensate preheater inlet	124.38	15.50	29.7	126.1	15.7
21	Feedwater from LP drum to boiler feed pumps	114.13	5.13	152.8	644.3	73.5
22	HP feedwater at BFP discharge to HP economiser	85.96	180.15	155.9	668.4	57.5
23	IP feedwater at BFP interstage to IP economiser	26.48	40.07	153.4	649.2	17.2
24	IP economiser water to performance fuel gas heater	11.00	38.26	240.7	1,041.0	11.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	22.2	90.3	920.4
29	HP attemperation spray water (BFP discharge)	1.39	180.15	155.9	668.4	0.9
30	Reheat attemperation spray water (BFP interstage)	0.30	40.07	153.4	649.2	0.2
31	Fuel gas to the station preheater burners (water-bath heaters)	0.03	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	340.00	MW
ST output (generator terminals)	175.05	MW
Gross output	515.05	MW
Auxiliary loads	-9.05	MW
Transformer losses (GSU, UAT, MV/LV)	-1.49	MW
NET OUTPUT (HV terminals)	504.51	MW
Heat input (LHV)	827.33	MW
Gross heat rate (LHV)	5,774	kJ/kWh
NET HEAT RATE (LHV)	5,903	kJ/kWh
Net efficiency (LHV)	60.98	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	85,636	Sm3/h
Circulating water flow	36,112	m3/h
GT load	80.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	82.18	%

CHECKS (guarantees, limits, balances)

OK energy balance CMIX: residual -0.002 MW (tolerance 0.25 MW)
OK energy balance COND: residual -0.081 MW (tolerance 0.93 MW)
OK energy balance FGH: residual +0.002 MW (tolerance 0.25 MW)
OK energy balance GT: residual +0.005 MW (tolerance 0.83 MW)
OK energy balance HRSG: residual +1.789 MW vs stated casing loss 1.768 MW (tolerance 0.85 MW)
OK energy balance PIPE: residual +0.342 MW (tolerance 1.00 MW)
OK energy balance ST: residual +0.003 MW (tolerance 0.72 MW)
OK gas flow 85,636 Sm3/h = 71.4 % of DP-OWNR-0033 (120,000 Sm3/h)
OK net export 504.5 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-GT80
CASE	Site Reference Conditions, natural gas, GT at 80 % load
DOCUMENT	ALP-HMB-SRC-NG-GT80
STATUS	preliminary - generated by engine hmb v1.1.1
DATA	b6f4b950c7f3b7e0 \| 2026-09-29
NOTE	all values calculated from the database; do not edit